

Annexure A

Short Answer :-

1). How does computer vision enable computers to interpret and understand visual information?

Computer vision enables computers to analyze images and videos by extracting useful information, recognizing objects, and understanding scenes similar to human vision.

a). Main Objectives of computer vision :

- *) Image understanding and interpretation
- *) Object detection and recognition
- *) Feature extraction
- *) scene analysis and decision-making

b). Two real-world applications :

- *) Medical image analysis (x-ray, MRI diagnosis)
- *) Autonomous vehicles.

2). Key features of computer vision for intelligent image analysis :

- *) Feature extraction
- *) Object detection and recognition
- *) Image segmentation
- *) pattern recognition.

a). Feature extraction :

Feature extraction is the process of identifying important characteristics of an image such as edges, shapes, colors, and textures.

b). Object recognition :

Object recognition helps computer vision systems identify and classify objects in images, improving accuracy and understanding.

3). Different levels of image processing in computer vision :

The levels are:

- * Low-level processing
- * Mid-level processing
- * High-level processing

a). Low-level image processing :

It involves basic operations like image enhancement, noise removal, and filtering without understanding image content.

b). Difference between mid-level and high-level processing:

Mid-level

Extracts features and segments images
Detects edges, objects, regions.

High level

understands and interprets images
performs recognition decision-making.

4). Digital image representation and components:

A digital image is represented as a matrix of pixels, where each pixel contains intensity or color information.

a). pixel :-

A pixel is the smallest unit of a digital image that represents a specific color or intensity value.

b). Effect of image resolution:

Higher resolution provides more details and better image quality, while lower resolution reduces clarity.

5). Image Sensing and acquisition :

Image sensing and acquisition convert real-world scenes into digital images using sensors that capture light and convert it into electrical signals.

a). Role of image sensor:-

An image sensor captures light from a scene and converts it into digital signals.

b). Two image acquisition devices:-

- * Digital camera

- * Scanner.

6). ~~Sampling and quantization~~

~~Sampling~~ and quantization convert continuous images into digital form.

a). Sampling:-

Sampling is the process of converting a continuous image into a grid of pixels.

6). Purpose of quantization :

Quantization converts pixel intensity values into discrete digital levels.

7). Image formation model :

The image formation model explains how light from object is captured by a camera and converted into a digital image.

a). Perspective projection :-

Perspective projection is the process of mapping 3D objects onto a 2D image plane.

b). Role of illumination :-

Illumination provides light that affects image brightness, color and visibility.

8). Image processing and computer vision relationship :

Image processing is the technique of enhancing, restoring and analyzing digital images.

b). Two difference:-

* Image processing focuses on image improvement; computer vision focuses on image understanding.

* Image processing produces enhanced images; computer vision produces information or decisions.

a). computer vision applications:

a). Healthcare applications:-

computer vision helps doctors analyze medical images like x-rays, CT scans, and MRI scans for disease detection.

b). Autonomous vehicle application:-

computer vision allows self-driving cars to detect roads, pedestrians, traffic signs, and obstacles.

105. Digital image fundamentals in computer vision:

a). Why pixels are basic building blocks:
Pixels contain the smallest units of visual information needed to form a digital image.

b). Influence of sampling and resolution:

- * Higher sampling and resolution improve image details and accuracy.

- * Lower resolution may cause loss of important information during analysis.

